Supplementary material

A Model parameterization and algebraic properties

Residue representations from the frozen ESM-2 model have dimension 1,280. Independent linear projections map wild-type, local-change, and cross-site-change features to the selected projection dimension. Amino-acid states use one-hot encodings. Both heads contain hidden layers of dimensions 256 and 128, GELU activations, and a scalar output. The pair head additionally receives calibrated single effects and observation indicators. Its sequence relation contains the logarithm of one plus the site gap and the gap divided by sequence length. Source-validation measurements select projection dimensions, learning rates, auxiliary weights, and ω.

The cross-site context is the change at one site after the other site is substituted,

$$\chi_{i \leftarrow j}(b) = h_i(x_p(j, b)) - h_i(s_p). \tag{11}$$

Encoding each single-mutant sequence once supplies residue states for every candidate pair involving that mutation. This gives the pair network mutation-dependent context without encoding double-mutant sequences. Sequence separation describes the endpoint relation; the structural variant adds reference Cα distances and residue confidence.

The normalized phenotype map is

$$g(z) = \frac{z + \sum_{k=1}^{20} b_k[\tanh(c_k z + d_k) - \tanh(d_k)]}{1 + \sum_{k=1}^{20} b_k c_k \operatorname{sech}^2(d_k)}, \qquad b_k, c_k > 0. \tag{12}$$

Softplus ensures positive parameters. Substitution of $z = 0$ gives $g(0) = 0$, and differentiation gives $g'(0) = 1$. The derivative is positive for every z, with the linear term preserving nonzero slopes in the tails. A positive factor τ sets the phenotype scale.

For $u = \widetilde{a}_i(a)$ and $v = \widetilde{a}_j(b)$, the predicted phenotype contrast decomposes as

$$\widehat{\epsilon}_{\text{global}} = \tau[g(u + v) - g(u) - g(v)], \tag{13}$$

$$\widehat{\epsilon}_{\text{pair}} = \tau[g(u + v + r_{ij}) - g(u + v)]. \tag{14}$$

Their sum is $\widehat{\epsilon}_p$. With the identity map and unit scale, the phenotype contrast equals r_{ij}. With a nonlinear map, the complete contrast includes both contributions.

Symmetry of F carries over to the cycle subtraction. At $b = w_j$, the first two terms cancel and so do the last two, giving $r_{ij}(a, w_j) = 0$. The other reference constraint follows identically. An endpoint term $v_i(a)$ contributes $v_i(a) - v_i(a) - v_i(w_i) + v_i(w_i) = 0$ to the subtraction. The second endpoint term and the constant cancel in the same way. Endpoint exchange and reference-state outputs are also checked on real data.

Matched controls for the pair function

For input removal, every evaluation of F receives model-predicted single effects $a_i(a), a_j(b)$ and zero observation indicators. External calibration still follows Equation (3) and the original label-permission rule. Measured values and availability indicators continue to calibrate that sum. Neither channel enters the pair network, whose input dimensions and parameter count remain unchanged.

The direct-head control predicts $r_{ij}(a, b) = F(i, a; j, b)$ with the same symmetric network. Its projections, endpoint features, measured inputs, hidden dimensions, and parameter count match SCEPI. This intervention isolates cycle subtraction while retaining endpoint symmetry and the external single-effect calibration.

Both controls reuse the full model's source-selected hyperparameters in each outer partition. Each repetition matches the pretrained single checkpoint, random initialization, source rows, sampler, optimizer, objectives, and checkpoint-selection rule. Maximum epochs and early-stopping patience are identical, while realized stopping epochs and fitted weights may differ. The label-mixing weight follows the same source-validation procedure.

Evaluation holds proteins, candidate pools, measurement margins, budgets, and aggregation fixed. Repeating the comparisons independently under the stricter homology grouping tests the predictive contributions of cycle subtraction and measured pair inputs across both groupings.

B Repetitions, aggregation, and statistical analysis

Five outer partitions define the 30% grouping. A source-trained model uses one partition for testing, the next for validation, and the remaining three for training. Training seeds 11, 23, 47, 89, and 131 give all 25 outer-fold and training-seed evaluations per model. Every eligible target protein is evaluated once per seed in its assigned test partition, using all eligible target double mutants. The resulting paired protein-repetition records number 795 for phenotype prediction, 775 for residual prediction, and 600 for standard compensation selection. The 0.5 kcal/mol margin cohort contains 575 records. Each protein's five repetitions are averaged before computing the equal-protein mean.

At each repetition, all compared methods use the same permitted target labels and candidate definition. Means use the paired cohort. Phenotype and residual comparisons cover 159 and 155 proteins, respectively, from 51 homology groups. Standard compensation covers 120 proteins and 44 groups. The 0.5 kcal/mol margin comparison covers 115 proteins and 43 groups. Source-validation configurations are retained across training repetitions. Candidate yield is excluded from source hyperparameter selection. Five outer partitions describe source-data separation; five training seeds describe repeated model fitting.

Uncertainty uses 10,000 homology-group bootstrap draws with statistical seed 20261013. Each draw samples groups with replacement and retains all member proteins. Its effect is the equal-protein mean over the sampled groups. Repeated model fits are averaged within protein before resampling. Comparisons with the same group membership use the same resampling draws. A bootstrap positive fraction is the fraction of resampled mean differences above zero. It describes uncertainty in the observed paired mean under this resampling scheme. Tables 4 and 5 report the top-10 endpoint. Table 6 reports the matched pair-function controls.

Table 4. Paired native top-10 hit-rate advantage of SCEPI. Gains and 95% bootstrap intervals are in percentage points. Positive fractions refer to resampled mean gains. Every comparison includes 120 proteins, 44 groups, and five model repetitions.

Comparator	Mean gain	95% interval	Positive fraction
Additive	4.83	[1.97, 7.95]	0.9991
Global nonlinear	6.77	[3.10, 10.11]	1.0000
ThermoMPNN-D	4.15	[−0.32, 7.08]	0.9628
Kermut	5.82	[2.59, 8.79]	0.9996
StabilityArc + Kermut	2.47	[−1.17, 5.55]	0.9165
Mean MLP	1.15	[−2.90, 3.76]	0.7387
Mean ridge	−0.20	[−4.10, 1.92]	0.4312

Table 5. Paired top-10 hit-rate advantage with the same measured single-effect sum at margin $\geq$ 0.5 kcal/mol. Gains and 95% bootstrap intervals are in percentage points. Every comparison includes 115 proteins, 43 groups, and five model repetitions.

Comparator	Mean gain	95% interval
Additive	6.63	[2.79, 9.96]
Global nonlinear	6.33	[2.38, 9.39]
ThermoMPNN-D	5.88	[2.39, 8.55]
Kermut	5.86	[2.19, 9.15]
StabilityArc + Kermut	3.03	[−0.64, 6.41]
Mean MLP	3.84	[0.27, 5.76]
Mean ridge	3.67	[0.77, 6.13]

Table 6. Paired advantage of SCEPI over matched pair-function controls under the 30% and 25% identity groupings. Residual RMSE reductions are in kcal/mol; native top-10 hit-rate gains are in percentage points. Brackets give 95% homology-group bootstrap intervals. Residual and selection comparisons contain 155 and 120 proteins, respectively. Each protein has five matched repetitions.

Grouping	Pair-function control	Residual reduction	Hit-rate gain
30%	Predicted singles	0.172 [0.152, 0.211]	3.03 [0.72, 4.32]
30%	Direct residual	0.041 [−0.014, 0.073]	0.52 [−3.04, 2.46]
25%	Predicted singles	0.210 [0.155, 0.263]	3.00 [−0.06, 4.36]
25%	Direct residual	0.026 [−0.019, 0.042]	−0.23 [−1.58, 0.81]

Measured pair inputs improve both mean residual error and native hit rate in all five repetitions under each grouping. Their residual comparison has 143 protein-level wins under the primary grouping and 142 under the stricter grouping, out of 155 proteins. The reference contrast lowers mean residual error in all five repetitions at each threshold, with 106 and 97 protein-level wins. Its native hit-rate gain is positive in four repetitions under the 30% grouping and two under the 25% grouping.

C Source model selection and implementation

Complete source cycles provide the residual target in Equation (1). For source proteins P, the residual objective gives equal weight to proteins and site pairs,

$$\mathcal{L}_\epsilon = \frac{1}{|P|}\sum_{p\in P}\frac{1}{|J_p|}\sum_{(i,j)\in J_p}\frac{1}{|Q_{pij}|}\sum_{q\in Q_{pij}}\mathrm{Huber}\left(\frac{\widehat{\epsilon}_q-\epsilon_q}{s}\right), \tag{15}$$

where J_p contains measured site pairs and Q_{pij} contains their cycles. The stability scale is $s = 1$ kcal/mol. Sampling across proteins, then pairs, then cycles implements this weighting.

The fitted objective is $\mathcal{L}_\epsilon + \lambda_y \mathcal{L}_y$. The observation term $\mathcal{L}_y$ is the negative log-likelihood for the predicted phenotype. Point observations use a Gaussian density; interval observations use the corresponding Gaussian interval probability. Each original observation appears once in this term. This criterion jointly fits the single head, pair function, and phenotype map. Residual-only and phenotype-only models are trained separately.

The single-effect model is fitted to source single-mutant measurements before joint training. The joint configuration grid contains projection dimensions of 64 and 128, learning rates of 10^{-4}, 3×10^{-4}, and 10^{-3}, and auxiliary phenotype weights of 0.1 and 1. The resulting twelve configurations are fitted separately in each outer partition. AdamW uses weight decay 10^{-4} and batches of 1,024 records. Training runs for at most 100 epochs, with early stopping after ten epochs without validation improvement. The residual loss uses Huber threshold 1 on the kcal/mol scale.

Single-label availability varies during training. Labels are hidden with probability 0.5; the model receives 128 labels with probability 0.25 and all single labels with probability 0.25. Source validation selects $\omega \in \{0, 0.5, 1\}$ in the 128-label condition. Model selection uses weighted residual RMSE, assigning weights 0.5 to zero labels, 0.25 to 128 labels, and 0.25 to all single labels. The selected configuration is refitted for each training repetition. Phenotype-only, residual-only, and representation variants have separate fitted parameters. For the additive model, validation uses phenotype error because its four-term residual is zero by construction.

Frozen sequence representations are cached by reference and single-mutant sequence. The pair predictor reuses those caches across double mutants. The structural variant adds Cα distance and residue confidence from the reference structure. Geometry permutations preserve the recorded distance values while changing their pair assignments. Inference interventions use the fitted structural model. Retrained interventions also use the changed assignments during fitting. Each comparison retains the same outer partition and information condition.

ThermoMPNN-D is retrained on source proteins with its double-mutant head and the published single-mutant data augmentation [3]. Its additive counterpart is trained separately. Kermut fits the permitted

target labels with the full composite kernel. Mean ridge and mean MLP fit those target labels using mean-pooled residue representations. ESM-2 additive and context scores provide fixed pretrained representation controls. The main external comparison evaluates each method's native predictions. Equation (10) replaces each method's single-effect sum with the same measured sum.

D StabilityArc architecture and target calibration

The StabilityArc comparison uses frozen ESMC-600M wild-type residue embeddings and attention-derived contacts [14]. Contact scores use symmetrization, average product correction (APC), and attention entropy weighting. Its decoder has four layers, hidden dimension 256, and eight attention heads with rotary position encoding. The full amino-acid embedding, sequence-distance features, and symmetric contact-aware pair residual are retained. Each of the ten source partitions is trained with the five fixed training seeds, giving 50 source models. Every source model excludes its target homology groups from both fitting and validation.

Each source model is trained for 40 epochs with AdamW, learning rate 10^{-4}, and weight decay 10^{-3}. Each assay contributes at most 40,000 measured variants per epoch. The objective combines Huber loss with 1,024 measured ranking pairs per assay. The Huber threshold is 1, and the two losses have equal weights. Double-mutant regression targets use the source-assay single-mutant median and median absolute deviation. Ranking uses measured stability order. Checkpoints are selected by source-validation double-mutant Spearman correlation. The public release supplies the architecture and a pooled checkpoint; the matched source-partition models are retrained here. Equal loss weights are an explicit implementation choice because the release does not specify them.

Target calibration uses the full Kermut covariance and Gaussian likelihood [6]. The retrained StabilityArc score replaces the prior mean input. All remaining kernel inputs and permitted target labels match the Kermut comparison. Target fitting uses 150 AdamW steps with learning rate 0.1 and weight decay 0.01. Only the permitted reference and single-mutant point labels are supplied. The posterior scores every eligible target double mutant and its constituent singles. The model's four-prediction contrast defines its residual.

E Component comparison details

Table 7. Top-10 compensation hit rates with the same measured single-effect sum. Each method contributes its own predicted four-term residual. The standard and margin cohorts contain 120 and 115 proteins, respectively.

Residual source	Standard pool (%)	Margin ≥ 0.5 (%)
SCEPI	**11.27**	**11.15**
ThermoMPNN-D	5.30	5.27
Kermut	5.47	5.29
StabilityArc + Kermut	8.17	8.12
Global nonlinear	4.70	4.82
Additive	4.42	4.52
Mean MLP	7.17	7.30
Mean ridge	7.33	7.48

Table 8. Full-target S-all results for predictors and model variants. RMSE and recovery are in kcal/mol. Hit rates are native top-10 compensation percentages. The phenotype, residual, and selection cohorts contain 159, 155, and 120 proteins. Each predictor is evaluated across five outer partitions and five training repetitions.

Predictor or variant	Phenotype	Residual	Hit rate	Recovery
SCEPI	0.657	0.595	11.22	−0.257
Additive	0.893	1.091	6.38	−0.387
Global nonlinear	0.911	0.635	4.45	−0.437
ThermoMPNN-D	0.777	0.805	7.07	−0.440
ThermoMPNN additive	0.967	1.091	5.30	−0.511
Kermut	0.798	0.847	5.40	−0.458
StabilityArc + Kermut	0.815	0.853	8.75	−0.344
Mean MLP	1.405	0.966	10.07	−0.413
Mean ridge	1.328	1.002	11.42	−0.354
ESM-2 additive	1.173	1.091	2.50	−0.794
ESM-2 context	1.107	0.864	8.50	−0.499
Pair removed	1.151	1.072	4.42	−0.432
Direct residual head	0.700	0.635	10.70	−0.302
Predicted singles	0.810	0.767	8.18	−0.396
Local context	0.637	0.575	11.47	−0.258
Wild-type context	0.643	0.573	10.53	−0.273
Identity features	0.617	0.557	9.12	−0.290
Phenotype loss	0.619	0.562	11.95	−0.229
Residual loss	1.071	0.686	8.72	−0.334
Frozen singles	0.689	0.619	9.75	−0.282
Identity map	0.651	0.593	11.13	−0.257
Structure	0.644	0.583	10.82	−0.263
Permuted distances^{I}	0.648	0.586	10.95	−0.261
Permuted distances^{R}	0.650	0.590	10.53	−0.270
Rewired distances^{I}	0.646	0.585	10.78	−0.264
Rewired distances^{R}	0.633	0.572	10.47	−0.257

Pair removal keeps the trained single-effect predictions and phenotype map fixed. The predicted-singles control replaces measured inputs inside F and retains measured single-effect calibration outside F. Local context omits cross-site changes. Wild-type context uses reference-sequence representations. Identity features use amino-acid identities and single effects, with hidden width adjusted to match the main model's parameter count. The remaining variants change the training loss, freeze the single-effect head, replace the phenotype map with the identity, or add structural distances. Superscripts I and R denote distance interventions during inference and retraining, respectively. Retrained additive and global nonlinear models optimize their own parameters.

Without cross-site context, residual RMSE is 0.575 kcal/mol and native hit rate is 11.47% under the primary grouping. Using amino-acid identities and single effects gives 0.557 kcal/mol and 9.12%, and using an identity phenotype map gives 0.593 kcal/mol and 11.13%.

Phenotype-only training reaches residual RMSE 0.562 kcal/mol and hit rate 11.95% under the primary grouping. It also improves both metrics under the stricter grouping, lowering residual RMSE from 0.604 to 0.584 kcal/mol and raising hit rate from 11.15% to 11.98%.

F Sensitivity to the homology threshold

At 25% sequence identity and the same coverage requirement, the phenotype and residual cohorts contain 45 groups. Compensation covers 39 groups, with 38 in the 0.5 kcal/mol margin subset. Protein membership

remains unchanged. Every predictor and component is evaluated across five outer partitions and five training repetitions (Table 9).

SCEPI retains residual RMSE of 0.604 kcal/mol under this grouping. ThermoMPNN-D reaches 0.771, Kermut 0.847, StabilityArc + Kermut 0.874, and global nonlinearity 0.650. Native compensation hit rates are 11.15% for SCEPI and 6.97%, 5.40%, and 8.23% for the three external methods, respectively. Mean selected recovery exceeds StabilityArc + Kermut by 0.092 kcal/mol.

With the same measured single-effect sum, top-10 hit rates are 11.17% for SCEPI, 4.42% for additivity, 4.63% for global nonlinearity, 5.68% for ThermoMPNN-D, 5.47% for Kermut, 8.57% for mean MLP, and 5.83% for mean ridge. At margin 0.5 kcal/mol, the corresponding rates are 11.23%, 4.52%, 4.75%, 5.65%, 5.29%, 8.77%, and 5.83%. StabilityArc + Kermut reaches 8.02% with the fixed sum and 7.95% at margin 0.5 kcal/mol. These paired comparisons retain the same proteins and label permissions. SCEPI's average selection advantage persists under both homology thresholds.

For the matched direct residual head, residual RMSE is 0.630 kcal/mol and native hit rate is 11.38%. SCEPI reaches 0.604 kcal/mol and 11.15%. Removing measured pair inputs while keeping external calibration gives 0.814 kcal/mol and 8.15%. With the same measured single-effect sum, the direct head reaches an 11.45% hit rate, the removed-input control 8.15%, and SCEPI 11.17%. Cycle subtraction retains its residual-accuracy benefit; measured pair inputs improve both prediction and selection.

Table 9. Full-target S-all sensitivity comparison at 25% sequence identity. The columns use the same units and protein cohorts as Table 8. Each predictor is evaluated across five outer partitions and five training repetitions.

Predictor or variant	Phenotype	Residual	Hit rate	Recovery
SCEPI	0.677	0.604	11.15	−0.256
Additive	0.994	1.091	6.65	−0.393
Global nonlinear	0.938	0.650	4.42	−0.432
ThermoMPNN-D	0.758	0.771	6.97	−0.432
ThermoMPNN additive	0.986	1.091	4.95	−0.536
Kermut	0.798	0.847	5.40	−0.458
StabilityArc + Kermut	0.840	0.874	8.23	−0.348
Mean MLP	1.211	0.922	11.20	−0.400
Mean ridge	1.638	1.039	11.25	−0.393
ESM-2 additive	1.173	1.091	2.50	−0.794
ESM-2 context	1.104	0.864	8.50	−0.499
Pair removed	1.151	1.073	4.42	−0.432
Direct residual head	0.699	0.630	11.38	−0.257
Predicted singles	0.855	0.814	8.15	−0.386
Local context	0.664	0.587	11.07	−0.266
Wild-type context	0.655	0.588	10.12	−0.294
Identity features	0.622	0.554	9.82	−0.287
Phenotype loss	0.653	0.584	11.98	−0.235
Residual loss	1.166	0.682	9.77	−0.311
Frozen singles	0.684	0.611	11.32	−0.256
Identity map	0.680	0.609	11.07	−0.252
Structure	0.683	0.609	11.75	−0.240
Permuted distances^{I}	0.685	0.611	11.93	−0.241
Permuted distances^{R}	0.677	0.606	11.75	−0.238
Rewired distances^{I}	0.684	0.609	11.62	−0.243
Rewired distances^{R}	0.688	0.615	10.62	−0.257

G Evaluation on the designated site-pair subset

The separate site-pair evaluation uses the designated unseen-site-pair test subset within each target protein. It intersects this subset with the same held-out homology groups. Site-pair split seeds are 11, 23, 47, 89, and 131. Candidate eligibility can change across these splits. The residual comparison contains 70 proteins, 34 groups, and 339 paired protein-repetition records. Compensation contains 63 proteins, 32 groups, and 269 records. Table 10 reports this subset without combining its values with the full-target results.

Table 10. Repeated S-all evaluation on the designated unseen-site-pair subset within held-out homology groups. Each eligible protein is averaged over its available repetitions.

Predictor	Residual RMSE	Native hit rate (%)	Recovery
SCEPI	0.549	5.75	−0.442
ThermoMPNN-D	0.778	4.88	−0.511
Kermut	0.833	3.29	−0.562
Global nonlinear	0.590	3.56	−0.528
Additive	1.080	4.71	−0.493

With the same measured single-effect sum, top-10 hit rates on this subset are 5.78% for SCEPI, 3.29% for ThermoMPNN-D, 3.07% for Kermut, 3.91% for global nonlinearity, and 3.62% for additivity. At margin 0.5 kcal/mol, the corresponding rates are 5.38%, 3.12%, 2.67%, 3.64%, and 3.34%. That margin cohort contains 62 proteins, 32 groups, and 263 paired protein-repetition records. Homology-group resampling with 10,000 draws and seed 20261010 gives an additive-comparison gain of 2.04 percentage points and a 95% interval of [0.67, 3.31]. Its gain over global nonlinearity is 1.74 points with an interval of [0.49, 3.12]. In S-all, every target double label is withheld from fitting. The site-pair subset defines an additional evaluation population. The target-combination-adaptation experiments use the pair split to test extrapolation from permitted training combinations.

H Target-label budgets and unseen-site-pair prediction

Partial target single-mutant measurements improve the transferred predictor (Table 11). Across the same 155 residual-evaluation proteins, RMSE is 0.793 kcal/mol in Z, 0.788 in S32, 0.772 in S128, and 0.595 in S-all. Only the target information condition varies. Source training and validation exclude each target homology group, and evaluation uses all eligible target double mutants. Every condition retains 155 proteins and 775 protein-repetition records. Kermut requires target measurements, so Z lists source-trained predictors. The separate target combination-adaptation experiments appear in Tables 13–15.

Table 11. Residual RMSE in kcal/mol at different target single-mutant budgets under 30% homology grouping. Each row covers the same 155 proteins and five training repetitions. Z supplies no target labels; S-all supplies all permitted target single-mutant labels.

Condition	SCEPI	Additive	Global nonlinear	ThermoMPNN-D	Kermut
Z	0.793	1.091	0.861	1.006	—
S32	0.788	1.091	0.857	0.943	0.998
S128	0.772	1.091	0.841	0.908	0.928
S-all	0.595	1.091	0.635	0.805	0.847

ESM-MSR is evaluated using the epoch-10 source-validation checkpoint from family30, fold 0, and repetition seed 11. All methods share the held-out proteins in that fold: 61 for residual prediction and 65 for

phenotype prediction. Table 12 separates Z and S-all. This one-fold, one-repetition comparison is distinct from the complete five-fold, five-repetition evaluation.

Table 12. Matched comparison with the available epoch-10 ESM-MSR checkpoint. RMSE uses kcal/mol. Columns report residual RMSE, phenotype RMSE, and phenotype Spearman correlation. Every method shares the same 61 residual proteins and 65 phenotype proteins within each information condition. Kermut requires target labels and is omitted in Z.

	Z			S-all		
Predictor	Residual	Phenotype	Spearman	Residual	Phenotype	Spearman
SCEPI	**0.744**	**1.071**	0.598	**0.550**	**0.621**	**0.767**
ESM-MSR	0.899	1.185	**0.646**	0.891	1.091	0.678
ThermoMPNN-D	1.031	1.237	0.587	0.805	0.772	0.707
Global nonlinear	0.829	1.100	0.534	0.632	0.888	0.735
Additive	1.035	1.136	0.588	1.035	0.885	0.731
Kermut	—	—	—	0.829	0.793	0.731

For the C conditions, SCEPI adapts to permitted target training combinations. Source initialization uses an identity phenotype map and holds the single-effect head fixed during source pair fitting. Target adaptation keeps this head fixed and updates the pair network and observation noise, with stopping selected by target validation. The target-only variant fits its single-effect head without supervised source weights. Both variants have the same target single-label permissions. Pair removal sets the fitted pair output to zero at inference. These adaptation models are evaluated separately from the jointly trained S-all predictor.

Superscripts S and T denote source initialization and target-only fitting. For mean ridge and mean MLP, they denote a source prior and target fitting. Numerical superscripts on MoCHI and MAVE-NN specify interaction order.

Each budget includes proteins with enough permitted training combinations and complete test cycles. Every comparator in a table uses the same protein-repetition intersection. Unseen-site-pair evaluation excludes all test pairs from target combination fitting. Random splitting shares site pairs between training and test combinations. The auxiliary two-pair cohort retains its designated pair split. Coverage reports proteins and matched protein-repetition records. Metrics average repetitions within protein before equal-protein averaging.

Table 13. Unseen-site-pair residual RMSE in kcal/mol after target combination adaptation. Coverage lists proteins and matched protein-repetition records. All methods share the cohort at each budget. Superscripts S and T denote source initialization and target fitting; numerical superscripts give interaction order. C-all uses all permitted training combinations. Test site pairs remain excluded from fitting.

Predictor	C128	C512	C2048	C-all
Coverage	70 / 339	45 / 219	4 / 17	70 / 339
SCEPI adaptation^{S}	0.567	0.570	0.424	0.559
Pair removed^{S}	1.080	1.163	0.967	1.080
SCEPI adaptation^{T}	0.799	0.788	0.495	0.774
Pair removed^{T}	1.080	1.163	0.967	1.080
SCEPI structure^{S}	**0.558**	**0.563**	**0.408**	**0.551**
SCEPI structure^{T}	0.795	0.785	0.467	0.772
ThermoMPNN-D	0.775	0.787	0.618	0.771
ThermoMPNN additive	1.080	1.163	0.967	1.080
Kermut	0.758	0.770	0.577	0.740
MoCHI^{1}	0.857	0.946	0.834	0.900
MoCHI^{2}	0.859	0.935	0.845	0.895
Mean ridge^{T}	0.933	0.994	0.767	0.921
Mean MLP^{T}	0.666	0.635	0.430	0.620
Mean ridge^{S}	1.000	1.067	0.917	0.999
Mean MLP^{S}	0.849	0.882	0.670	0.816

Table 14. Random-split residual RMSE in kcal/mol after target combination adaptation. Every row shares the cohort and averaging rule within each budget. Superscripts follow Table 13. Random splitting evaluates new amino-acid combinations at site pairs represented in training.

Predictor	C128	C512	C2048	C-all
Coverage	125 / 601	90 / 450	11 / 55	125 / 601
SCEPI adaptation^{S}	0.355	0.315	0.279	0.277
Pair removed^{S}	1.065	1.122	1.212	1.065
SCEPI adaptation^{T}	0.401	0.358	0.311	0.311
Pair removed^{T}	1.065	1.122	1.212	1.065
SCEPI structure^{S}	**0.351**	**0.310**	**0.273**	**0.274**
SCEPI structure^{T}	0.399	0.350	0.303	0.304
ThermoMPNN-D	0.424	0.394	0.381	0.385
ThermoMPNN additive	1.065	1.122	1.212	1.065
Kermut	0.447	0.405	0.355	0.346
MoCHI^{1}	0.737	0.802	0.991	0.810
MoCHI^{2}	0.736	0.800	0.997	0.813
Mean ridge^{T}	0.527	0.498	0.568	0.449
Mean MLP^{T}	0.389	0.339	0.283	0.305
Mean ridge^{S}	0.971	1.031	1.124	0.967
Mean MLP^{S}	0.749	0.746	0.726	0.690

Table 15. Residual RMSE in kcal/mol on the auxiliary two-pair cohort. Coverage lists proteins and matched protein-repetition records. Superscripts follow Table 13.

Predictor	C128	C-all
Coverage	27 / 131	29 / 140
SCEPI adaptation^{S}	0.683	0.687
Pair removed^{S}	1.163	1.172
SCEPI adaptation^{T}	1.138	1.149
Pair removed^{T}	1.163	1.172
SCEPI structure^{S}	**0.653**	**0.659**
SCEPI structure^{T}	1.147	1.157
ThermoMPNN-D	0.813	0.823
ThermoMPNN additive	1.163	1.172
Kermut	0.840	0.827
Mean ridge^{S}	1.063	1.074
Mean MLP^{S}	1.002	1.001

I Functional assays and higher-order target adaptation

Six ProteinGym assays supply 74,598 functional double mutants on their original measurement scales [16]. Keeping the three KRAS abundance blocks separate gives eight conditions and 7,898 single mutants. Seven conditions provide measured wild-type values, yielding 70,198 complete cycles. HIS7 lacks an observed wild type and contributes phenotype-ranking tests. Each fit uses only its permitted measurement condition (Table 16).

We evaluate random combination splits and held-out site pairs at C128, C512, C2048, and C-all. Each condition supplies its permitted single mutants and training combinations. Test combinations remain excluded from fitting and validation. The C2048 condition excludes HIS7 because its training pool is smaller than the budget. Phenotype rank correlation is computed on measured test double mutants. Residual evaluation requires at least 30 complete test cycles. To compare assay scales, residual RMSE is divided by the population standard deviation of permitted training point values. Tables report equal-assay means. Repetitions are averaged within condition, the three KRAS conditions are averaged within their assay, and assays receive equal weight.

Table 16. Accepted functional double-mutant data and complete measured cycles. Each row retains its original measurement condition. The three KRAS conditions belong to one abundance assay. HIS7 lacks an observed wild type and contributes no complete cycles.

Assay condition	Singles	Doubles	Complete cycles
D7PM05_CLYGR fluorescence	1,169	10,148	9,517
GFP_AEQVI fluorescence	1,084	12,777	11,954
HIS7_YEAST selection fitness	168	1,475	0
Q6WV12_9MAXI fluorescence	1,141	15,992	15,717
Q8WTC7_9CNID fluorescence	1,201	11,260	10,822
KRAS abundance, block 1	1,188	11,946	11,904
KRAS abundance, block 2	1,122	7,345	7,184
KRAS abundance, block 3	825	3,655	3,100
Total	7,898	74,598	70,198

Functional adaptation separates single-effect fitting from pair fitting. The SCEPI variant fits single-effect predictions and a target-specific monotone phenotype map, freezes both, and then fits the pair function. Source initialization uses models trained with an identity phenotype map and a fixed single-effect head. The recorded homology relation excludes related source proteins, and eligible source models are averaged at the effect level. Target-only fitting uses no supervised source weights. Both variants supply predicted single effects to the pair network. They test source initialization and target combination fitting separately from measured-input conditioning in the primary S-all predictor.

The independent no-pair control fits the first stage without adding pair effects. Inference-time pair removal uses the fitted two-stage model and sets its pair output to zero. Learning rates are 10^{-4}, 3×10^{-4}, and 10^{-3}. Target-only models compare projection dimensions 64 and 128. Pair fitting compares observation-loss weights 0.1 and 1. Each stage runs for at most 100 epochs with ten-epoch validation patience. Target validation selects phenotype rank correlation, with identifiable phenotype RMSE breaking ties.

MoCHI and MAVE-NN compare first-order and second-order sequence models [4, 11]. Reference-free analysis (RFA) uses second-order effects with the author's sigmoid phenotype link [15]. The fixed configuration sets $\lambda = 0$. Its optimization budget is 500 L-BFGS iterations, with gradient threshold 10^{-5} and relative objective tolerance 10^{-4} over 25 iterations. Normalization has a separate 500-iteration cap. The validation-selected configuration compares twelve penalties, $\lambda/n \in \{0, 10^{-6}, 10^{-5.5}, \ldots, 10^{-1}\}$. Test outcomes are excluded from fitting and configuration selection.

RFA covers 350 eligible fits, comprising 310 functional-assay fits and 40 FLIP2 fits. Of these, 284 use the fixed configuration and 66 use validation-selected penalties. Thirty additional tasks lack the required sample budget or target training and validation labels and are excluded from fitting.

RFA gives constant GFP predictions in all five C128 repetitions under each split and in two unseen-pair C512 repetitions. Their twelve Spearman correlations are undefined. Rank comparisons use the same finite protein-repetition records for every method. C128 therefore excludes GFP and retains five assays, seven conditions, and 35 records. Unseen-pair C512 retains six assays, eight conditions, and 38 records. The other six-assay rank columns contain 40 records. C2048 retains five assays and 35 records after excluding HIS7. Residual evaluation retains all seven eligible conditions and 35 records at every budget, including the constant-prediction cases.

On unseen pairs at C128, the source-initialized interaction variant reaches normalized residual RMSE 0.508 (Table 17). Pair removal gives 0.551 and target-only interaction fitting 0.537. First-order MAVE-NN reaches 0.484. At C-all, RFA reaches 0.409 and the source-initialized variant 0.448.

Phenotype ranking favors pair removal at C128 (Table 18). On the full six-assay unseen-pair ranking cohort, the source-initialized variant has Spearman correlation 0.730, compared with 0.741 after pair removal. Pair fitting improves residual accuracy and reduces phenotype rank correlation in this setting.

Table 17. Normalized residual RMSE on functional double mutants. Every column contains five assays, seven measurement conditions, and five repetitions. Errors are divided by each condition's training phenotype standard deviation before equal-assay averaging. Superscripts S and T denote source initialization and target-only fitting; numerical superscripts give interaction order. C-all uses the permitted training combinations.

	Random combinations				Unseen site pairs			
Predictor	C128	C512	C2048	C-all	C128	C512	C2048	C-all
SCEPI adaptation^{S}	0.503	0.484	0.465	0.435	0.508	0.491	0.472	0.448
Pair removed^{S}	0.551	0.529	0.508	0.478	0.551	0.532	0.514	0.488
Independent no-pair^{S}	0.551	0.529	0.508	0.480	0.551	0.532	0.514	0.490
SCEPI adaptation^{T}	0.537	0.515	0.489	0.439	0.537	0.519	0.491	0.459
Pair removed^{T}	0.549	0.527	0.499	0.459	0.549	0.530	0.502	0.469
Independent no-pair^{T}	0.549	0.527	0.499	0.459	0.549	0.530	0.502	0.469
MoCHI^{1}	0.545	0.497	0.453	0.452	0.543	0.499	0.458	0.427
MoCHI^{2}	0.532	0.498	0.453	0.444	0.533	0.497	0.464	0.430
MAVE-NN^{1}	0.476	0.457	0.429	0.407	0.484	0.464	0.431	0.415
MAVE-NN^{2}	0.481	0.460	0.442	0.425	0.484	0.461	0.446	0.437
RFA	0.544	0.524	0.446	0.398	0.548	0.528	0.441	0.409
Mean ridge	0.749	0.687	0.633	0.576	0.792	0.716	0.651	0.610
Mean MLP	0.800	0.715	0.657	0.604	0.793	0.738	0.673	0.619

Table 18. Phenotype Spearman correlation on common finite records for functional double mutants. Every model has five fitting repetitions. Assay counts are given for each budget. Conditions are averaged within assay before equal-assay averaging. Superscripts follow Table 17.

	Random combinations				Unseen site pairs			
Predictor	C128	C512	C2048	C-all	C128	C512	C2048	C-all
Assays	5	6	5	6	5	6	5	6
SCEPI adaptation^{S}	0.739	0.731	0.828	0.771	0.748	0.737	0.827	0.763
Pair removed^{S}	0.754	0.734	0.830	0.767	0.758	0.742	0.829	0.767
Independent no-pair^{S}	0.755	0.734	0.830	0.769	0.755	0.745	0.830	0.767
SCEPI adaptation^{T}	0.716	0.719	0.824	0.766	0.731	0.734	0.824	0.769
Pair removed^{T}	0.720	0.723	0.823	0.767	0.733	0.733	0.824	0.769
Independent no-pair^{T}	0.720	0.723	0.823	0.766	0.732	0.733	0.824	0.769
MoCHI^{1}	0.706	0.711	0.824	0.772	0.707	0.718	0.833	0.760
MoCHI^{2}	0.713	0.727	0.836	0.773	0.721	0.738	0.844	0.766
MAVE-NN^{1}	0.726	0.731	0.830	0.775	0.753	0.741	0.838	0.769
MAVE-NN^{2}	0.726	0.732	0.831	0.769	0.751	0.741	0.840	0.763
RFA	0.705	0.692	0.791	0.747	0.716	0.692	0.798	0.739
Mean ridge	0.602	0.592	0.695	0.662	0.617	0.597	0.691	0.659
Mean MLP	0.557	0.548	0.662	0.649	0.566	0.555	0.661	0.641

FLIP2 evaluates TrpB one-to-many, two-to-many, and by-position splits; amylase one-to-many; IRED two-to-many; Hydro three-to-many; and three Hydro background-transfer splits [17]. We retain the continuous phenotypes and official test assignments. Each split is evaluated separately across five repetitions. Hydro three-to-many averages its three target backgrounds equally. The background-transfer tests provide training measurements from the other backgrounds. Per-background MoCHI, MAVE-NN, and RFA fits require

target-background training labels and are unavailable in those three tests. Pooled representation controls and the adaptation variants use the permitted cross-background training pool.

TrpB by-position correlation rises from 0.167 to 0.233 after source-initialized pair fitting (Tables 19 and 20). One-to-many correlation falls from 0.252 to −0.214. On Hydro three-to-many, the corresponding values are 0.356 and 0.201. The source-initialized independent no-pair model reaches 0.703 on amylase. Each split compares source initialization, target fitting, and inference-time pair removal for the predicted-single adaptation variant.

Table 19. FLIP2 phenotype Spearman correlation on TrpB, amylase, and IRED. All values use the official test split and average five repetitions. TrpB columns distinguish one-to-many, two-to-many, and by-position protocols. Amylase uses one-to-many; IRED uses two-to-many. Superscripts follow Table 17.

	TrpB			Amylase	IRED
Predictor	1-to-many	2-to-many	By position	1-to-many	2-to-many
SCEPI adaptation^{S}	−0.214	0.155	0.233	0.697	0.200
Pair removed^{S}	0.252	0.172	0.167	0.697	0.221
Independent no-pair^{S}	0.252	0.215	0.172	0.703	0.221
SCEPI adaptation^{T}	0.388	0.206	0.180	0.677	0.222
Pair removed^{T}	0.388	0.413	0.180	0.677	0.219
Independent no-pair^{T}	0.388	0.413	0.180	0.677	0.219
MoCHI^{1}	0.247	0.481	0.256	0.593	0.173
MoCHI^{2}	0.336	0.482	0.218	0.621	0.191
MAVE-NN^{1}	0.245	0.393	0.254	0.574	0.175
MAVE-NN^{2}	0.338	0.460	0.135	0.584	0.175
RFA	0.319	0.180	0.263	0.625	0.181
Mean ridge	0.454	0.466	0.283	0.693	0.199
Mean MLP	0.437	0.485	0.302	0.695	0.173

Table 20. FLIP2 Hydro phenotype Spearman correlation. Three-to-many averages three target backgrounds; each remaining column holds out the indicated background. Every value averages five repetitions. Dashes indicate local fits without target-background training labels. Superscripts follow Table 17.

Predictor	Three-to-many	P01053	P06241	P0A9X9
SCEPI adaptation^{S}	0.201	0.368	0.193	0.025
Pair removed^{S}	0.356	0.420	0.197	0.188
Independent no-pair^{S}	0.356	0.420	0.215	0.188
SCEPI adaptation^{T}	0.059	0.408	0.219	0.036
Pair removed^{T}	0.284	0.408	0.251	0.193
Independent no-pair^{T}	0.284	0.404	0.258	0.209
MoCHI^{1}	0.549	—	—	—
MoCHI^{2}	0.557	—	—	—
MAVE-NN^{1}	0.554	—	—	—
MAVE-NN^{2}	0.565	—	—	—
RFA	0.519	—	—	—
Mean ridge	0.586	0.239	0.025	0.168
Mean MLP	0.614	0.317	0.303	0.025

J Interpretation of the selection endpoint

Complete, point-valued double-mutant cycles define the retrospective evaluation cohort. Their double-mutant outcomes are excluded from model training and candidate scores. Measurement status and distance from assay limits define evaluation strata. The margin is the minimum of $y+1$ and $5-y$ across all four observations. A threshold of 0.5 kcal/mol excludes cycles close to either assay limit. This analysis conditions the evaluation on measured outcomes inside the assay range.

Recovery is $y_{AB} - \max(y_A, y_B)$. A hit requires at least 0.5 kcal/mol recovery, with each single losing at least 0.5 kcal/mol relative to wild type. Hit rate measures successful discoveries at a fixed budget; mean recovery measures continuous candidate quality. Meeting the recovery criterion does not require restoring wild-type stability.

General stability ranking is evaluated on a broader continuous pool of 65 proteins in one repetition with seed 11. With the same measured single-effect sum, top-10 mean stability changes differ from additive and global nonlinear ranking by -0.0018 and -0.0054 kcal/mol, respectively.

K Random selection and compensatory enrichment

For protein p, let N_p be the number of eligible candidates and H_p the number satisfying the measured recovery threshold. Uniform sampling of K candidates without replacement gives a hit count $X_p \sim \text{Hypergeometric}(N_p, H_p, K)$. Its expected hit rate is $q_p = H_p/N_p$, and

$$\text{Var}(X_p/K) = \frac{q_p(1-q_p)(N_p-K)}{K(N_p-1)}. \tag{16}$$

Expected mean recovery equals the candidate-pool mean. Both expectations are independent of K and are calculated exactly from all eligible candidates without random-ranking simulation.

Let $\overline{h}_p(K)$ be a predictor's hit rate averaged over its five fitted repetitions. For the same P eligible proteins, enrichment is

$$E(K) = \frac{|P|^{-1}\sum_{p\in P}\overline{h}_p(K)}{|P|^{-1}\sum_{p\in P} q_p}. \tag{17}$$

This ratio uses the same equal-protein aggregation as the primary results. It remains defined when individual proteins have no observed hits. The standard cohort contains 79,564 candidates and 1,718 hits across 120 proteins. Its pooled candidate hit fraction is 2.16%, and its equal-protein mean random hit rate is 2.63%. The latter is the denominator for every reported enrichment.

The random comparison uses the same fixed model predictions, candidate eligibility, recovery threshold, budgets, and measurement margins as the other selection comparisons. Each predictor's five repetitions are averaged within protein before aggregation. Uncertainty uses 10,000 paired homology-group bootstrap draws with seed 20261019. Each draw resamples model and random results together and recalculates both means and their ratio. These intervals quantify sampling uncertainty in the observed cohort. The exact random expectation has no Monte Carlo ranking error.

Table 21. SCEPI native selection relative to exact uniform random selection under 30% homology grouping. The same 120 proteins and 44 groups enter every budget. Random hit rate is 2.63%. Hit-rate gains are percentage points; recovery gains are kcal/mol. Brackets give 95% homology-group bootstrap intervals for enrichment.

Budget	SCEPI hit rate (%)	Hit-rate gain	Enrichment (×)	Recovery gain
1	14.50	11.87	5.52 [2.58, 8.24]	0.784
3	12.89	10.26	4.90 [2.59, 7.11]	0.728
10	11.22	8.59	4.27 [2.74, 5.89]	0.643
30	8.56	5.93	3.26 [2.43, 4.40]	0.529

At $K = 10$, SCEPI's hit-rate gain over random selection has a 95% interval of [4.01, 12.07] percentage points. Its recovery gain has an interval of [0.499, 0.720] kcal/mol. The stricter homology grouping gives 4.24-fold native enrichment. A margin of 0.5 kcal/mol retains 115 proteins and 75,017 candidates. Native enrichment is 4.15 and 4.19 under the 30% and 25% groupings. With the same measured single-effect sum, enrichment is 4.17 and 4.20 in these cohorts (Table 22).

Table 22. Top-10 random-selection sensitivity. Random rates and model rates are equal-protein means. The last two columns report SCEPI enrichment with native predictions and the fixed measured single-effect sum. Both use the same random expectation. The 30% grouping has 44/44/43 groups at the three margins; the 25% grouping has 39/39/38.

Grouping	Margin	Proteins	Random hit (%)	Native (×)	Fixed sum (×)
30%	0.00	120	2.63	4.27	4.29
30%	0.25	120	2.68	4.17	4.19
30%	0.50	115	2.68	4.15	4.17
25%	0.00	120	2.63	4.24	4.25
25%	0.25	120	2.68	4.17	4.17
25%	0.50	115	2.68	4.19	4.20